ATOSFleetManagement: Architecture and Development Guide

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1) Scope

ATOSFleetManagement is a lightweight ATOS variant for TruckObject development. This stack intentionally removes scenario engine and ISO22133 dependencies from runtime behavior.

```
- Included: TruckObjectControl, TruckObjectGUI, bridge stack
- Excluded: OpenScenarioGateway, JournalControl, EsminiAdapter, ObjectControl, ISO22133 runtime path
```

2) Runtime Graph

```
+-------------------+            +---------------------+
|  TruckObjectGUI   |<---------->|  Bridge (Fox/ROS)   |
+-------------------+            +---------------------+
          |
          v
+-----------------------------------------+
|           TruckObjectControl            |
|  sub: /atos/truck_objects/cot           |
|  pub: /atos/truck_objects/speed_command |
+-----------------------------------------+
          ^
          |
+-----------------------------------------+
| Physical TruckObjects on test track     |
+-----------------------------------------+
```

3) Placeholder Control Logic

```
Input COT topic: /atos/truck_objects/cot (std_msgs/String)
Expected payload: id=<truck_id>;distance_m=<value>;tcp_connected=<0|1>

Policy:
- Consider only trucks with valid TCP flag and fresh COT
- Compute minimum distance gap along trajectory among eligible trucks
- If gap < 200 m: publish target_speed_kmh=0
- Else if gap < 400 m: publish target_speed_kmh=30
- Else: no new speed restriction command

Output topic: /atos/truck_objects/speed_command (std_msgs/String)
```

4) Build and Run

```
source /opt/ros/humble/setup.bash
cd /home/sepast/Documents/repos/ATOS
./scripts/build_atosfleetmanagement.sh

source /opt/ros/humble/setup.bash
source /home/sepast/atos_ws/install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
```